

# Which language, and what that admits

A signal is read; a convention is declared; nothing is inferred. The whole design of this layer is that distinction.

## 1 • resolving the language

strongest signal first, stop at the first hit

### 1. declared

language on /v1/search

A deliberate statement about a corpus, made at READ time — so a vault ingested under one language answers correctly under another with no re-ingest.

Outranks everything below.

### 2. else the script

where a script settles it

Greek · Georgian · Hangul — one script, one language. A fact, not a reading.

Cyrillic → Russian, Devanagari → Hindi: the majority language, and labelled approximate because that is what it is.

### 3. else the drawer itself

language\_of\_drawer

A text carrying der, die, und, nicht is German. Only CLOSED-class words vote — a loanword does not get one.

Decisive or nothing: three hits and twice the runner-up, else stay silent.

Read per candidate, not per query: a vault may hold several languages, and the drawer is the unit that has one.

## 2 • five rules, every one of them pairwise

each answers about two strings and builds no class

### suffix\_family

SHAPE, not length.

-er is German-only:

Kind/Kinder, Haus/Häuser

### inflection\_family

A CLOSED set of endings

per language, stem ≥ 3:

libro/libri, книга/книги

### agglutinative\_family

Endings that stack:

Turkish, Korean, Georgian, Hungarian

### ar\_root\_family

144 roots × 20 patterns

= 2859 forms. Two words meet only if ONE root fits.

### IRREGULAR

~110 forms no rule

over shape can reach:

go/went, Mann/Männer

### Why not a stemmer

A stemmer builds an equivalence class, and one false friend poisons every member. Measured: Snowball Greek merges πολύ (much) with πόλη (city). A pairwise rule costs only its own pair.

### Why Arabic needed its own table

An ALLOWLIST: a form the table cannot generate matches nothing. Subsequence rules failed three times — بيت→بيوت and يجيب→يجب are the same string operation, and no rule over shape separates them.

## 3 • three channels, and only two of them are a claim about words

### lexical\_exact

The drawer said the word. admits

### lexical\_morph

It holds a word BUILT ON it. admits — and says so separately

### semantic

Cosine, above the gate the EMBEDDER itself calibrates.

### Measured — 191 paradigm pairs, 19 languages

|                                     |              |
|-------------------------------------|--------------|
| lexical recall, declared            | 55.0% → 100% |
| lexical recall, undeclared          | → 100%       |
| pairs resting on the embedder alone | 20 → 0       |
| negative controls                   | 0 → 48       |

### The prices, pinned rather than hidden

Declaring German is what makes -er legal, and -er also relates flow/flower, corn/corner, butt/butter. That is the trade, stated.

A recall measurement cannot justify a precision decision: adding -er moved the population figure by 0.21 links per query — invisible — and broke five negative controls. The controls are what caught it.